Arus Daryayi | عروس دریایی

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Victoria Hoang
Computation Arts & Computer Science
Concordia University
victoriatthoang@gmail.com

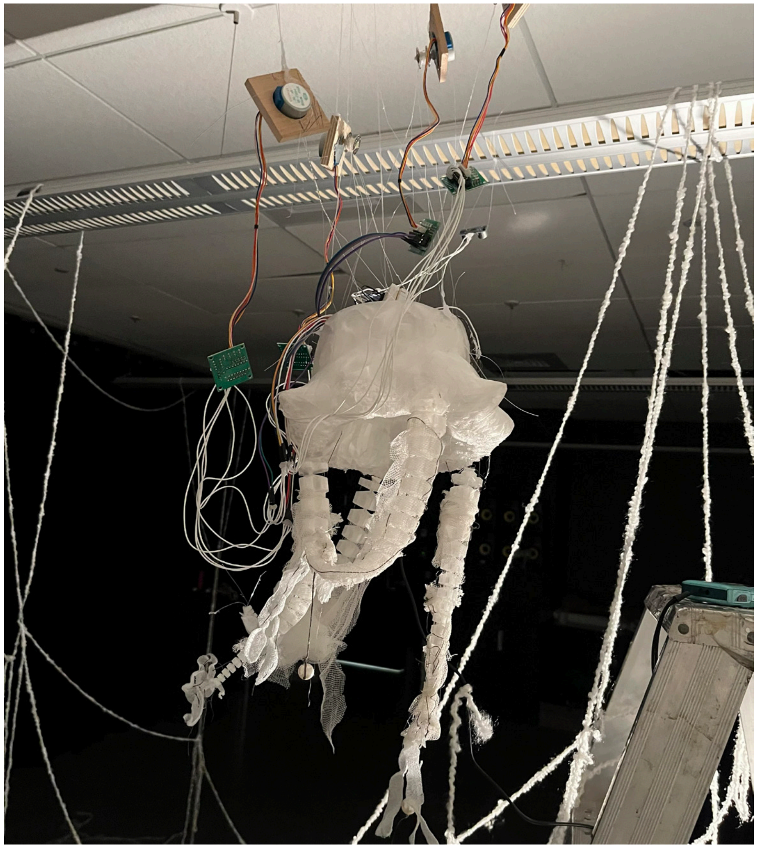
Kiana Rezaee
Computation Arts & Community Public Affairs and Policy
Concordia University
kia.kianpour6@gmail.com

ABSTRACT

Arus Daryayi is a research-creation installation featuring two jellyfish-like soft robotic creatures designed to challenge the dominant culture of instant gratification embedded in modern human-technology relationships. Developed over nine weeks and exhibited in Concordia University's Holodeck, the work invites participants into a slow, reciprocal dialogue with a non-human organism through touch, sound, and light. Drawing on cybernetic feedback, conductive thread, stepper motors, and nature-inspired form, the installation resists the opaque logic of the digital Black Box in favor of transparency, mechanical vulnerability, and organic presence. The parallel development of two distinct creature forms, one open and intuitive, the other sealed and resistant, served as both a design method and a living argument: that concealment creates friction, and that openness fosters empathy. Ultimately, the project turned its critique inward, implicating its own creators in the very culture of fast-paced production it sought to question. Arus Daryayi proposes interaction not as a service transaction but as a meditative and grounding encounter, asking participants and makers alike to slow down and meet the artifact at its own rhythm.

KEYWORDS

soft robotics · research-creation · interaction design · slow design · black box · open systems · cybernetics · tactile interaction · bioplastics



INTRODUCTION

Arus Daryayi is an installation-based interactive experience featuring two jelly-fish or octopus-like creatures, the implementation of the work features aspects of performance art. It is the result of a 9 week research-creation project. One creature is suspended from the ceiling and the other is left on the ground. This project was displayed in Concordia's department of Design and Computation Art's dark room, the Holodeck. Through sound, movement, and light the suspended creature increasingly encourages participants to engage through sight and touch. The piece explores the tension between the digital Black Box and organic based-design, both in its process and results. This project was a culmination of our own experience with attempting to create two different creatures.

Heavily inspired by Spirobs (Wang et Al. 2025), we utilized these open source 3d models of tentacle-like robotic arms. We hypothesized that designing for human intuition involves showcasing original references to nature and can increase empathy and interest in objects. In the context of our own project, empathy has grown to mean understanding and patience. By utilizing the direct imagery of ocean creatures as biological surrogates and anecdotes, the work invites participants to move away from the efficiency of consumer technology and toward a practice of tranquility.

Arus Daryayi full installation, Holodeck

1 Persian for jellyfish Lit. Sea bride

As we moved forward with the project we also increasingly applied this concept to our own relationship with the two creatures, the one with organic form was intuitively what created a sense of empathy within us. There is also of course the implicit push we have towards the designers of Spirobs and other nature inspired innovations to include their inspirations in a more explicit manner. Scholars such as Dicks (2024) argue that contemporary biomimicry often isolates natural functions from their aesthetic origins, a process that conceptualizes innovation and what it looks like as a separate entity of purely biomorphic design choice. All of this led us to make the piece relatively performance and environment based and not just object oriented, making sure to retain the aesthetics of both machinery and nature.

INTENTIONS

Arus Daryayi was initially ideated as a project to explore the tension of the traditional relationship between humans and technology through the creation of a soft robotic organism. The planned interaction strategy relied on a self-regulating cybernetic feedback loop where the presence and movements of the audience directly influences the behavior of the organism. By utilizing ultrasonic sensors to detect proximity and translating that data into soft robotic movements and real-time soundscapes the installation aimed to create a sense of non-human sentience.

This multisensory experience was to empower users to explore the artifact with curiosity and touch, challenging the anthropocentric view of digital devices as mere service providers. Ultimately the project aimed to transform the act of interaction into a meaningful conversation with an expressive living presence. However we would now define the central intention as an act of facilitating an intimate and playful dialogue that encourages participants to move at a slower pace and engage with the artifact at its own rhythm instead of getting instant gratification.

SIMILAR PROJECTS

The development of Arus Daryayi is situated within a broader lineage of research creation that utilizes soft robotics and biomimicry to redefine the boundaries between the mechanical and the organic. By examining existing works that prioritize emotional resonance, autonomous movement, and bio-inspired geometry, we can better understand how our project contributes to the discourse on non-human sentience and tactile engagement. The following projects served as vital benchmarks for our material choices and interactive goals.

[A] The Coral Morph by Xinyi Huang & Daniela M. Romano

Coral Morph is a moving textile installation that encourages emotional regulation and sensory engagement via soft robotics material movements. It includes touch sensors and heart-rate tracking to dynamically control inflatables. When participants either touch or breathe near the artifact, its main body as well as its surrounding tentacle-like tubes respond by changing shape and colour, seeking to elicit serenity. The installation was studied with 55 participants, who found it to be safe, interesting, and emotionally intelligent due to its tactile responses and organic movement (Huang et Romano 2025).

[B] In Love With the World by Anicka Yi

In Love With the World is a large-scale installation of aerobes that are bulbous robotic creatures resembling jellyfish that float autonomously in the space. It uses electronic tracking to respond to people and the overall environment and further engage smell via changing scentscapes. By combining organic form with robotics, this work seeks to question how machines could inhabit the world (Anicka Yi Studio 2021).

[C] Spirobs by Zhanchi Wang, Nikolas Ferris, and Xi Wei

SpiRobs are soft robots inspired by natural appendages like octopus arms and elephant trunks, which often follow a logarithmic spiral shape. The robots are 3D printed with flexible material and controlled by two or three cables that allow the body to curl and uncurl. This spiral movement lets the robot reach, wrap around, and grasp objects of different shapes and sizes using its flexible body. The design is simple, low-cost, and scalable, with versions ranging from tiny robots that can handle delicate organisms to larger robots capable of lifting heavy objects or operating on drones (Wang et Al. 2025).

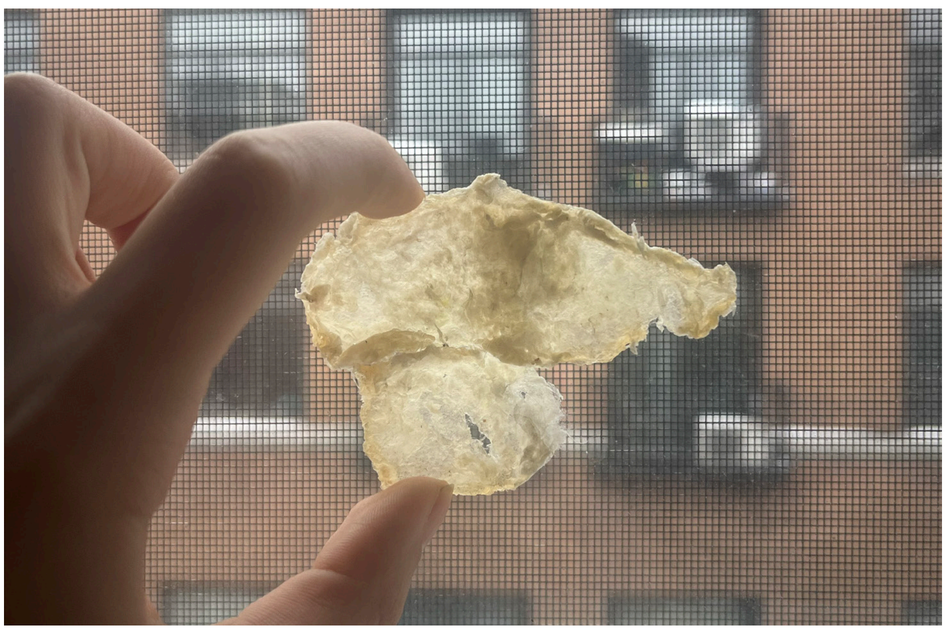
PROTOTYPING & ADVANCEMENTS

Materials

The initial research phase prioritized the use of gelatin based bioplastics to achieve a truly organic form. By exploring a material that is naturally biodegradable and transient the project sought to emphasize slowness, ocean-life, and the touchfeel of a living organism. However this material was ultimately abandoned due to factors such as unpredictable structural integrity (shrinkage due to dehydration) and loss of color (due to oxidation). This material experimentation was rooted in a desire to introduce more organic matter and counter the culture of fast-paced consumerism and the disposability of modern technological objects.

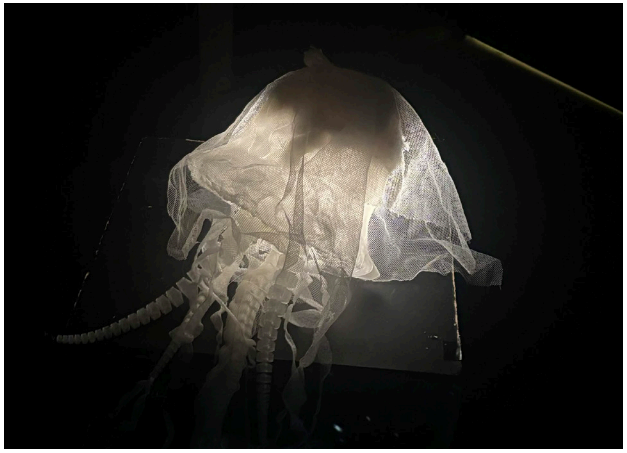
Because the research required to stabilize the bioplastics went beyond the project's scope we chose a 3D printed flexible plastic material, TPU, which our spiral legs were made of to ensure mechanical reliability while still exploring transparency and softness. By moving away from the rapid decay of the bioplastic the final work ensures that the interaction remains focused on the slow and gradual movement of the jellyfish rather than the immediate degradation of the material itself.

To summarize, we ultimately discovered how difficult it is to integrate biomaterial-based electronics even in a research-creation setting. The current climate of consumer based research prioritizes fast based production methods. Thus, we ended up partaking in our own hypothesis on technology by our lack of ability to slow down. We did continue to experiment with cloth and wool as other natural materials but ended up deciding to play into the synthetic plastic. We landed on using mesh fabric and conductive thread for the soft circuitry. We also began to think about creating a memetic environment for the jelly using similar cloth, plastic or foam like materials and even yarn. These materials were all thrifted.



Bioplastic sample (gelatin, water, glycerin) (before oxydization)

Bioplastic samples (gelatin, water, glycerin) (after oxydization)



Thrifted fabric samples

During the printing stage with Concordia's Digital Fabrication lab, we were inspired by two prints that came out completely different. The softest and most organic of these misprints only printed halfway while the only successful full print resulted in a dense structure with a rigid square infill. This served as an anecdotal shell for the concept of enchantment and the Black Box through several unique misprints. Working with these two distinct forms reflected the core issues of designing for the Black Box versus the open system during our process and the subsequent results. Our definition of the Black Box is related to enchantment as explored by Lupetti and Murray-Rust (2023) who argue that enchantment masks technical and environmental realities, trapping users and creators in a state of functional ignorance. Through our metaphor, we insinuate and argue that disenchanted technology that does not hide its mechanics can still be enchanting in other ways especially if it showcases its original morphology.

The organic halfway print proved easy to thread because the open top provided direct access to the interior, allowing for the installation of fishing wire for the legs and threaded soft sensors. In contrast the dense Black Box print was nearly impossible to pierce and eventually required being split in half at the side just to facilitate assembly, it was easy to split in half due to the fact it was falling apart from the pattern of its infill. In this model the fishing wire remains visible and hanging in the middle of the two halves, further emphasizing the struggle to conceal internal mechanisms within a closed form.

This technical friction serves as a physical manifestation of the project goals as it highlights the accessibility of open systems against the stubborn and opaque nature of traditional technological enclosures. We found that this was also apparent in our exploration of soldering our circuitry. The more we attempted to hide within our materials, the more issues began to arise with how detailed our design had to be.

Black Box Jellyfish (denser TPU print)

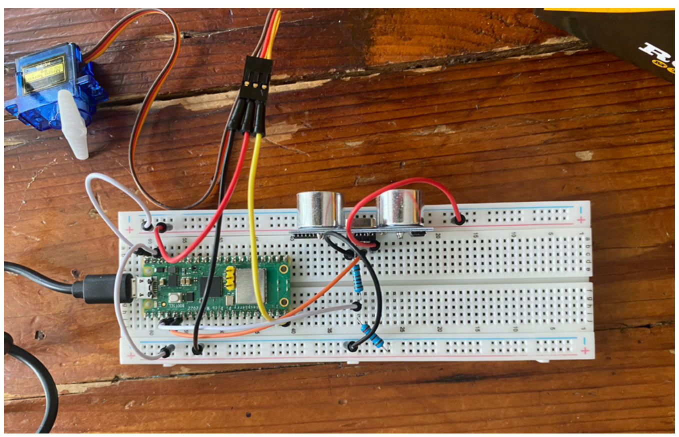
Open Jellyfish (less dense TPU print)

Circuit & Code

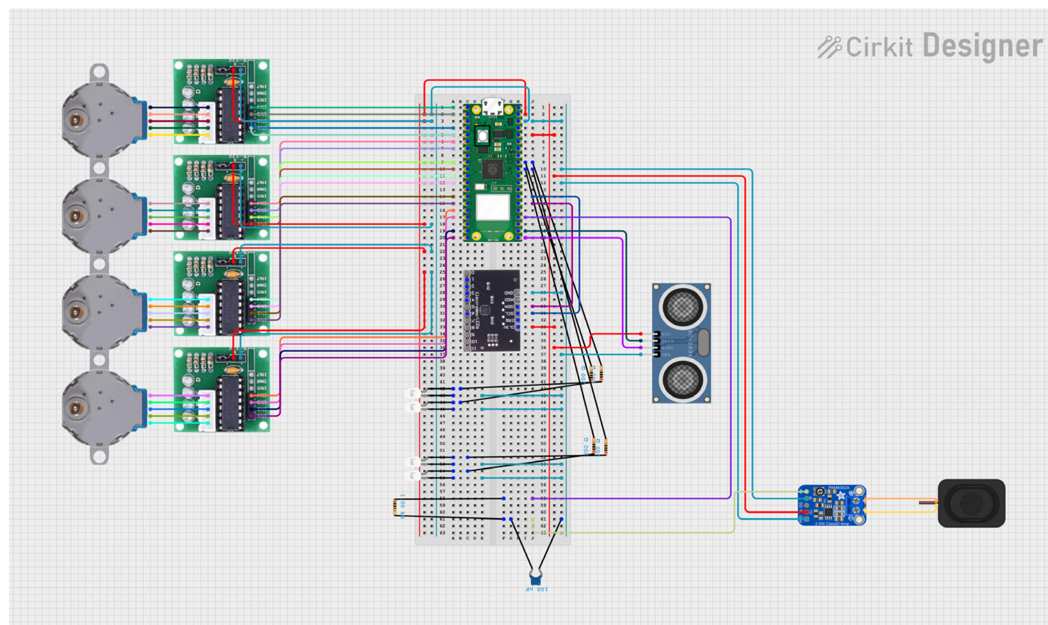
The initial development phase utilized a basic distance sensor and servo motor for its circuit. During testing in early March, the system lacked the mechanical torque and precision required to articulate the tentacles or spin easily. The movement was jittery and failed to convey the intended organic sentience due to the type of motor and its limited rotation amount. Due to these issues, we ultimately decided to use stepper motors as they have good rotation control and torque. The motors were the most essential part of the changes made, in terms of figuring out mechanics. The tension required to move them and the way the fishing wire would get tangled made it so that the easiest position to set up was with the project already suspended.

We also changed the main interaction to a touch sensor using an MPR121 and conductive thread as this allowed for the touch aspect that we were interested in. While we initially wanted the legs to curl instantly and rapidly in response to touch, the inherent speed of the stepper motors led us to shift our focus from instant gratification to prioritizing slowness. Thus, if a user were to release a leg before it would finish curling, the leg would unfurl, pushing users to spend more time with the artifact. The HC-SR04 ultrasonic sensor ended up remaining as a way for us to control lights within the object, increasing the speed of fading in and out as users approached the installation.

Several routines were explored for the motor sequencing. Initially, the circuit used 3 motors and all 12 electrode pins of the MPR121, where each motor was mapped to 4 touch electrodes. This initial logic had a single string from each leg connected to one motor, such that each motor controlled one direction for all four legs at a time. However, this setup decentralized the touch interaction, as touching one leg would lead to all four legs curling, and was rather reliant on users touching different sides of the leg to exhibit different reactions. As this did not align with our goals, we shifted to using four stepper motors: 1 motor controlled the inward motion for all 4 legs, 1 motor controlled the sideways motion for all 4 legs, 1 motor controlled the outward motion for 2 legs, and 1 motor controlled the outward motion for the other 2 legs.



Initial circuit
Servo motor & HC-SR04



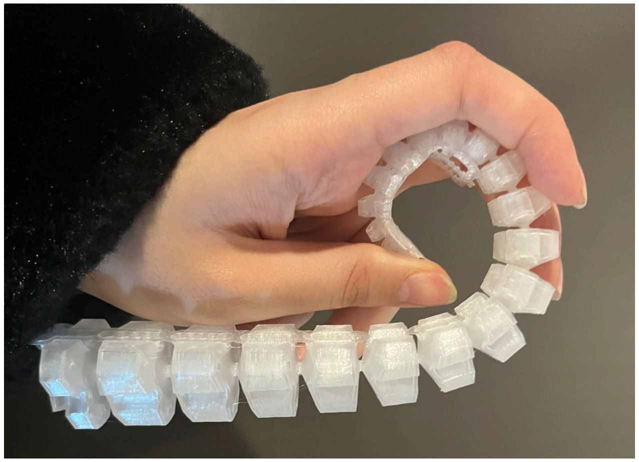
Fnal circuit
Stepper motors & MPR121
HC-SR04 & LEDs
Amp & Speaker with RC filter

Motor Controls

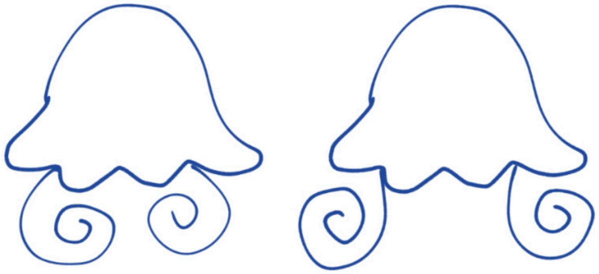
The motors would follow the following routine in round-robin, until a touch sensor was activated:

Motor 1: all 4 legs outward
Motor 4: all 4 legs inward
Motor 2: 2 legs sideways
Motor 3: 2 legs sideways

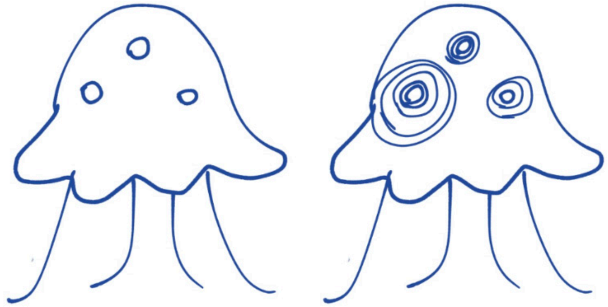
If a touch sensor was activated, this sequence would be interrupted. All legs will firstly return to home, determined by the motor's default starting position before any rotation. Depending on which leg was touched, motor 2 or 3 will then rotate outward so long as the user continues touching the leg. If the user releases the leg, the motor will rotate backwards to home, uncurling. Once all legs are homed (no touches sensed), the sequence will restart from the beginning.



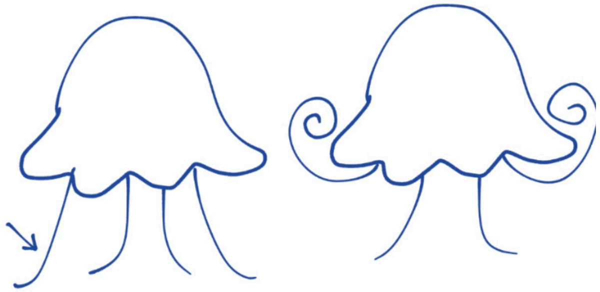
Movement & Light Language



Breathing and Welcoming:
Controlled by motor three and four. A rhythmic grouping of four legs moving in and out to simulate a calm, living state.



Bioluminescence:
When in the presence of humans as detected by the distance sensor, the creature's internal lights fade in and out



Joy of Touch:
Controlled by motor one and two. Full extension of a pair of opposite legs trying to wrap around hands when a participant is touching the conductive thread on the legs. This signals a positive interaction.

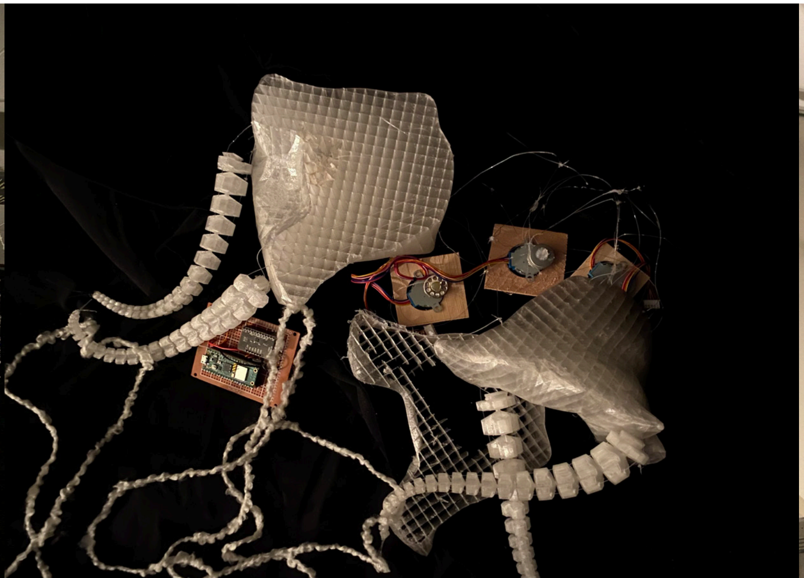
INTERACTION DESIGN & NARRATIVE

The interaction design of Arus Daryayi is not traditional, meaning there is no command and control structure. According to Dubberly et Al. (2009), The interaction system of Arus Daryayi would be a 2-2 Conversing model, moving beyond the typical 0-2 relationship where a human learns a linear machine. By rejecting this, the jellyfish becomes a participant in an I/you-referenced dialogue. Rather than viewing the participant or creator as a user triggering a function, the system is programmed to facilitate a reciprocal effect between the human and the non-human organism. The motors and fishing wire were the biggest part of this during our research-creation process. By also utilizing conductive thread, the interaction becomes a form of tactile provocation where the creature's response is a slow frequency movement that mirrors the weight and pacing of a physical caress and light haptics.

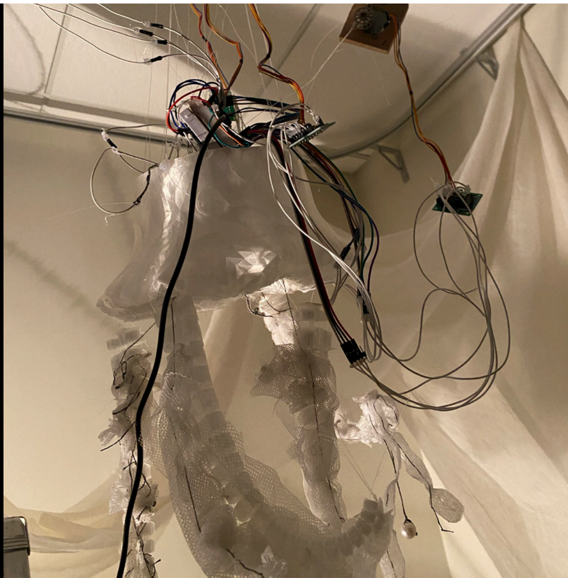
Visually, the environment draws the eye of the audience to three participants, the interactor next to or atop a ladder, the black-box iteration connected by yarn tendrils on the ground, and the successful iteration suspended in a central spot of the environment. The successful creature is most reactive when touch is detected, as there is a shift to the perceptive feedback loop. This conversational mode is achieved through a reciprocal effect; the human must learn the creature's movement language while the creature responds to tactile provocations with its own kinetic goals. This system establishes a second loop where the pace of interaction is a negotiated agreement between the two.

[1] Interactor atop ladder

[2] Black-box Jellyfish



[3] Open Jellyfish



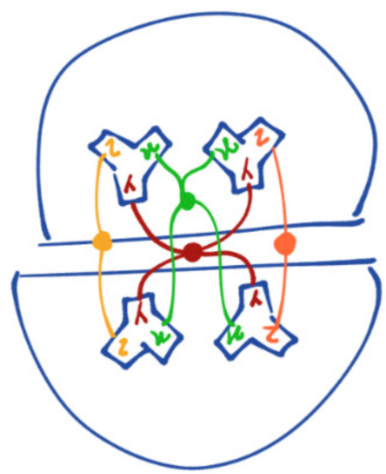
As legs wrap toward the hand, the participant feels and observes the tension of the fishing wire motors and tendons, making the creature's internal effort visible and felt. The kinetic intentionality transforms the interaction from a simple one to one trigger into a meaningful conversation, challenging the instant gratification and hidden aspects of modern interfaces. This transparency fosters a deeper technical empathy because the friction of the machine is no longer hidden but is instead part of the dialogue. By rejecting the opaque service oriented nature of the digital black box, Arus Daryayi's interaction design facilitates an intimate and playful encounter that asks the participant to move at the rhythm of the artifact, reclaiming the act of interaction as a meditative and grounding experience.

OBSERVATIONS & DISCOVERIES

We would divide our discovery and observational process in two parts, one for the open creature and the other for the black box creature.

However some general trends were noticed, primarily the biggest technical challenge throughout the development process was the management of motor tension. Because the movement relies on a cable driven system, the tangledness of the fishing wire created inconsistent tension across the four legs. This made the upward movement of the tentacles particularly difficult to achieve reliably.

Furthermore, our own documentation of these mechanical failures was often incomplete because we found it difficult to remain patient with the slow capture required by the subject. During the final exhibition, we observed that we could not invite participants to spend significant time with the work. This suggests that the human intuition for instant gratification is difficult to override, even when the visual language of the object explicitly calls for a slower pace especially as creators in the research context ourselves.



Schematic of the wiring and why it was not possible to detangle

The Black Box Creature

A significant discovery during development was the physical planning and design difficulty of the Black Box model. Concealing the internal systems required complex maneuvering and forced the use of difficult piercing methods to route fishing wire through the structure. Additionally, the design of our movements did not permit for us to have easy wiring in the blackbox version as it was impossible to detangle, since it only really had two pierced holes on each half.

Our hypothesis on designing for the black box and fast paced production came to fruition with our attempt at soldering a circuit to put inside the black box jelly to hide its circuitry. This resulted in a completely non functioning circuit which we attribute to the lack of time to explore more problem solving. This technical failure served as a physical manifestation of our research. By attempting to force the electronics into a closed and hidden form under a strict deadline, we engaged in the very culture of fast production that we intended to critique. The black box effectively locked us out of our own creation, making it impossible to repair or adjust the system once it was sealed.
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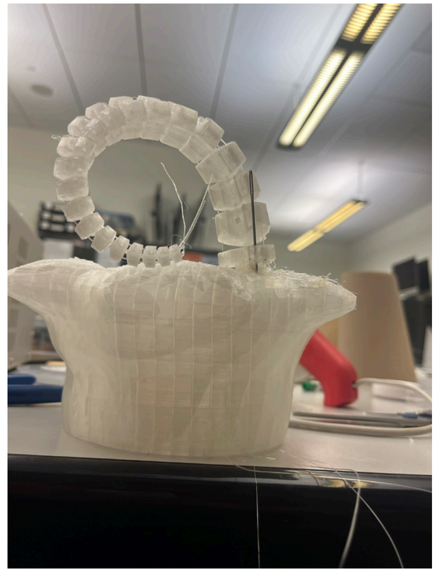

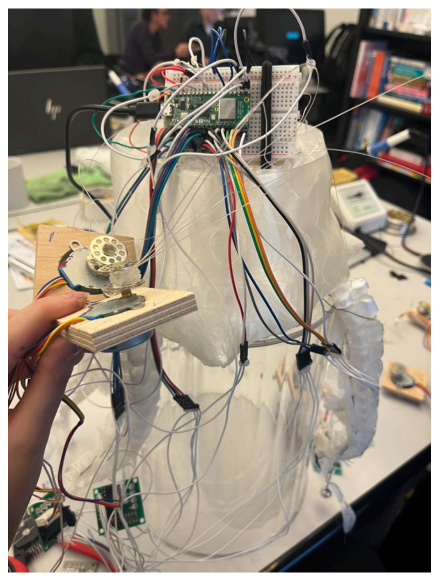

The Open Jellyfish

While the open jellyfish still had the same tangling issue, the process proved far more intuitive to assemble once suspended as the low infill allowed us to maneuver the strings to not touch each other. This was especially important due to the addition of touch sensors, as we could visually confirm that the conductive threads were not crossing or shorting against the motor housing. The semi-exposed structure allowed for easier mechanical troubleshooting and established a more honest relationship with the creator.

Unlike the black box, the open model allowed the mechanical nerves and tendons of the creature to remain part of the aesthetic. We discovered that this transparency actually enhanced the sense of empathy, as the participant could see the physical effort of the motors as they pulled the strings. This model did not rely on the illusion of a perfect digital object but instead embraced its own mechanical vulnerability. By allowing the internal workings to be legible, we created an artifact that invites the participant to understand its rhythm rather than just consume its output. This confirmed that an open system is far more conducive to the practice of tranquility and maintenance than a sealed consumer enclosure.

As creators, we found ourselves gravitating toward the open jellyfish throughout the entire fabrication process. There was a profound sense of relief in working with a form that did not fight against our touch or hide its needs behind a rigid shell. While the black box felt like a source of constant frustration and technical dead ends, the open model invited a more fluid and meditative way of making. It allowed us to move at a pace that felt natural to the materials themselves rather than the artificial speed of a production deadline. By being able to see and reach the internal mechanism, we felt less like operators of a machine. This personal preference ultimately confirmed that the transparency we sought to provide for the participant was equally necessary for our own creative well being, making the open system the more successful realization of our intentions. We reflected this in our installation and performance.

Fidelity to Proposal

The realization of Arus Daryayi maintained a degree of fidelity to the initial conceptual goals of exploring non-human sentience and soft robotic interaction, though the technical execution underwent a significant evolution from the March 10th prototype. Our initial proposal prioritized the use of organic biomaterials and simple servo motors to create movement. However, as the project progressed, we had to balance our desire for materiality with the mechanical requirements of a cable-driven system.

The project remained meaningful in soft robotics but developed another core intention of challenging the black box and gratification. The unexpected discovery of the halfway misprint actually took the project beyond its initial scope, providing a physical metaphor for open-source systems that we had not originally planned. This allowed us to situate the work within a broader critique of consumer technology, where the friction of the open jellyfish became a celebrated feature rather than a bug.

FUTURE IMPROVEMENTS

Looking forward, there are several avenues for refinement that would deepen the immersion of Arus Daryayi. Technically, the most immediate improvement would involve the development of a custom PCB or a more robust mounting system for the electronics, this would ensure the safety of all of its components. While the openness of the current wiring was a conceptual success, it remains physically vulnerable to long-term exhibition wear. Additionally making a custom wire management system for the fishing wire and the conductive thread would be a huge improvement and make the project more robust.

From a material standpoint, we hope to revisit the integration of bioplastics by developing a hybrid structure consisting of a TPU skeleton for mechanical reliability. This would then be stitched over with a stabilized gelatin to achieve the transient, organic aesthetic we initially researched. Additionally, future iterations could include sound as we originally intended. We would also like to explore having the jellies “talk” to each other. Finally, expanding the motor sequencing to allow for more granular, individual leg control would enable a more complex dance between the two jellies, creating a truly synchronized, multisensory performance.

REFERENCES

Anicka Yi Studio. “In Love With The World.” Accessed April 17, 2026. https://anickayi.com/exhibitions/in-love-with-the-world. (2021).

Dicks, Henry. “Contemporary Biomimicry and the Neglect of Aesthetics.” Techne: Research in Philosophy & Technology 28, no. 3 (2024).

Dubberly, H., U. Haque, and P. Pangaro. What is interaction? Are there different types?(2009). 2012.

Huang, Xinyi, and Daniela M. Romano. “Coral morph: an artistic shape-changing textile installation for mindful emotion regulation in the Wild.” International Journal of Human–Computer Interaction 41, no. 2 (2025): 1173-1189.

Lupetti, Maria Luce, and Dave Murray-Rust. “(Un) making AI magic: A design taxonomy.” In Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems, pp. 1-21. 2024.

Wang, Zhanchi, Nikolaos M. Freris, and Xi Wei. “SpiRobs: Logarithmic spiral-shaped robots for versatile grasping across scales.” Device 3, no. 4 (2025).